TRANSISTOR LABS

FM-1 emulator

User Manual - Version 1.5.4

Free AU / VST3 / CLAP plugin for macOS, Windows and Linux

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1. Introduction

FM-1 emulator runs the firmware of the M-VAVE FM-1 pocket synth inside your DAW. It emulates the FM-1's processor, screen, keys, knobs, audio and MIDI, so firmwares behave as they do on the real device. You play it from your DAW's MIDI or with the mouse on its front panel.

FM-1 emulator is free. It is an unofficial product and is not affiliated with or endorsed by M-VAVE.

2. System Requirements

- **macOS:** macOS 12 Monterey or later, Mac with Apple silicon (M1 or later). AU, VST3 and CLAP.
- **Windows:** Windows 10 or 11, 64-bit (x64). VST3 and CLAP.
- **Linux:** 64-bit x86 distribution with X11 (e.g. Ubuntu 22.04 or later). VST3 and CLAP.
- A host that loads AU, VST3 or CLAP instruments (Logic, GarageBand, Ableton Live, Cubase, Reaper, Bitwig, FL Studio, Studio One...).
- An FM-1 firmware file (.fwsc). Firmware is not included: see chapter 4.

3. Installation

macOS: double-click *FM-1-emulator-1.5.4-macos.pkg* and follow the installer. It installs the AU, VST3 and CLAP into /Library/Audio/Plug-Ins.

macOS security warning. The installer is not notarised by Apple, so the first time you open it macOS says it cannot be opened because it is from an unidentified developer. To open it:

- Right-click (or Control-click) the .pkg and choose **Open**, then click **Open** again in the dialog.
- Or: try to open it once, then go to **System Preferences > Security & Privacy > General** (macOS 12) or **System Settings > Privacy & Security** (macOS 13 and later) and click **Open Anyway**.

Windows: run *FM-1-emulator-1.5.4-windows-x64-setup.exe*. It installs FM-1.vst3 into Common Files\VST3 and FM-1.clap into Common Files\CLAP. The VST3 needs the CLAP file to be installed too, so keep both. If Windows SmartScreen warns you, click **More info > Run anyway**.

Linux: unpack the .tar.gz and run *./install.sh*. It copies the plugins into ~/.vst3 and ~/.clap.

Plugins only: the -plugins archives contain the plugin files to copy by hand into your plugin folders.

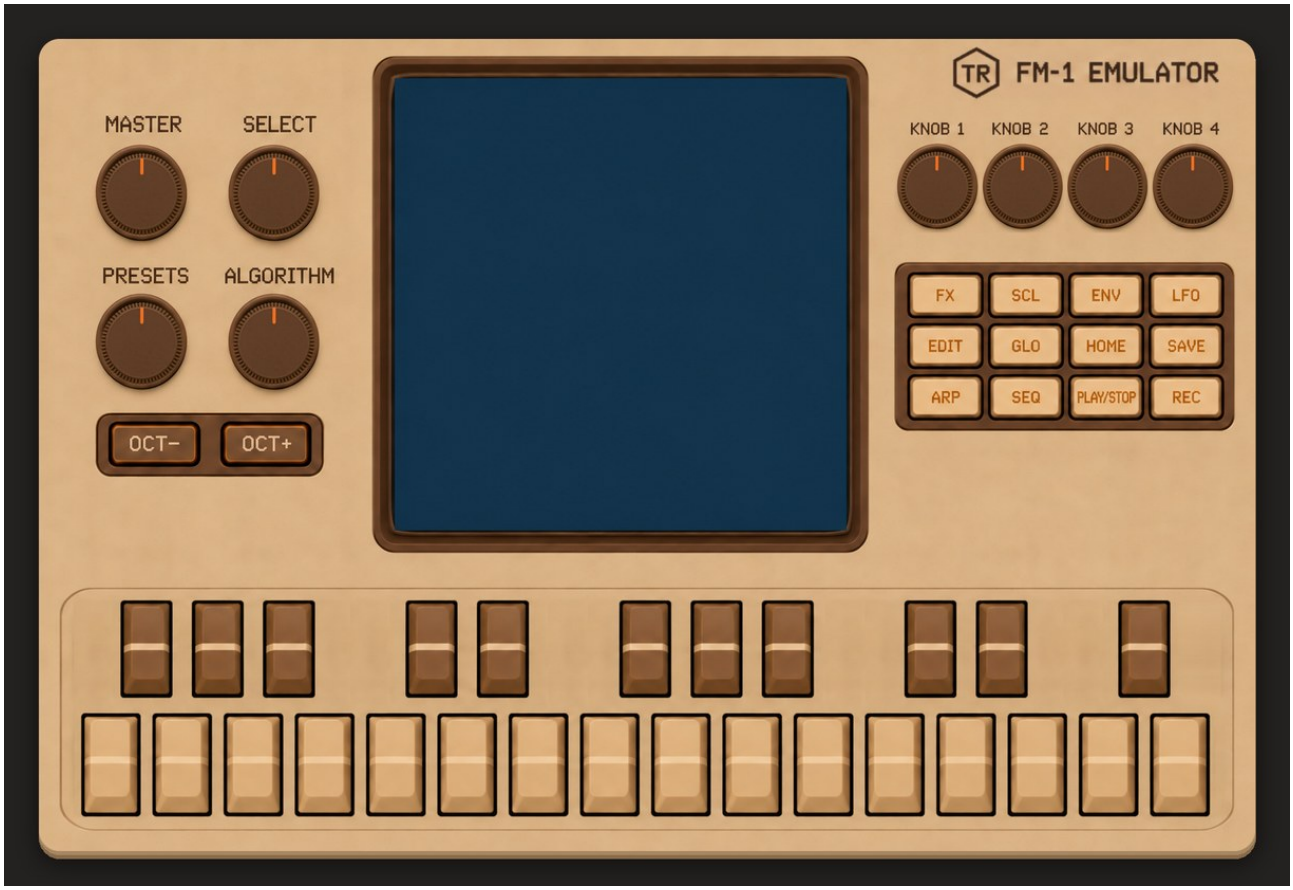
Restart your DAW after installing. The plugin is listed as **FM-1 emulator** by **Transistor Labs**.

4. Getting a Firmware

The plugin needs an FM-1 firmware package (.fwsc) to make sound. Firmware is copyrighted by its authors and is not bundled. Community firmwares are available from their authors' websites; a list with links is on the fm1-emu project page (https://joakimeriksson.github.io/fm1-emu/). Download a .fwsc file and keep it somewhere you can find it.

Once loaded, the firmware is stored inside your DAW project, together with the patches it saves, so the project opens with the same sound next time.

5. The Plugin Window



At the top of the window is the control bar; below it is the FM-1 panel: MASTER, SELECT, PRESETS and ALGORITHM knobs and OCT-/OCT+ on the left, the large screen in the middle, the logo, KNOB 1-4 and the twelve function buttons (three rows of four) on the right and the 27-key keyboard (F3 to G5) at the bottom. Keys and buttons glow orange when the firmware lights their LEDs.

- **Load firmware / save...** opens a .fwsc firmware package or a shared .fm1save setup.
- **Export save...** saves the running firmware, its patches and settings as a .fm1save file you can share.
- **MIDI channel** moves all incoming notes to one channel. Most firmwares listen on a single channel (M-VAVE's own on 2): if your MIDI keyboard plays nothing while the panel keys do, set this to the firmware's channel.
- **Sync to host** (on by default) sends your DAW's tempo to the firmware as MIDI clock, with Start and Stop following the transport, so arpeggiators and sequencers play in time. It works with firmwares that follow MIDI clock.
- **CPU clock** sets the emulated chip's speed (240 MHz like the real one). Lower it if your computer can't keep up; raise it if a firmware runs out of time. Changing it restarts the firmware.
- **Lead** sets how far ahead of the host the firmware runs. Smaller is quicker to respond; go back up if "frames late" appears in the status bar.
- The **status** text shows the firmware name, speed and any late frames.
- **About** shows the version and credits.
- **Window size:** choose 75%, 90%, 100%, 110% or 125% from the **Size** menu at the left of the top bar, next to **About** (the current size has a tick), then close and reopen the plugin window: it opens

at the new size. The size is saved in your DAW project. (A plugin window resizing itself while open is not followed reliably by every host, so the size changes when the window opens.) The screen scales with the panel: shown small it is smoothed; shown larger than its real 240 x 240 pixels it switches to sharp pixels, like a real LCD up close.

6. Playing

- Play from your DAW's MIDI track or a MIDI keyboard.
- On the panel, click a key to play it; drag across the keys for a glissando.
- Drag a knob up or down, or use the mouse wheel over it.
- MASTER is the host parameter *Master*, so it can be automated.
- Buttons work while held down, as on the hardware.
- Notes from your MIDI keyboard or DAW clips press the matching panel keys too (notes outside F3-G5 show an octave up or down).

An SMK-37 firmware package shows the SMK-37 panel instead, with its pads, faders and wheels.

7. Syncing to Your DAW's Tempo

FM-1 emulator always sends your DAW's tempo to the firmware as MIDI clock (24 pulses a beat), through the FM-1's MIDI IN (TRS) jack, whenever **Sync to host** is on. It also sends **Start** when the DAW starts playing and **Stop** when it stops. Whether the firmware follows that clock is up to the firmware: most keep their own tempo until you set them to listen to an external clock. The steps are the same for every firmware; only the name and place of its setting differ.

Step by step:

- **1. Turn Sync to host on** in the plugin's top bar (it is on by default).
- **2. Find the firmware's clock setting.** Look in its global, system or settings menu (often reached with GLO, EDIT or HOME). It is usually called **SYNC**, **CLOCK**, **CLK SRC**, **CLOCK IN** or **TEMPO SOURCE**. The firmware's own manual or web page says where it is.
- **3. Set it to an external clock.** Choose the option named **TRS**, **MIDI**, **EXT**, **External** or **Slave**. If it offers both **USB** and **TRS**, choose **TRS**: that is the input the plugin sends to. (**INT** or **Internal** means the firmware's own tempo.)
- **4. Save it** if the firmware asks, so the setting stays. FM-1 emulator keeps the firmware's settings in your DAW project.
- **5. Press Play in the DAW.** The firmware starts with it and follows its tempo; change the DAW's tempo and the firmware follows. Stop in the DAW stops it.
- **6. Check:** the tempo on the firmware's screen should match the DAW's. If it does not move, see below.

Things to know:

- Leave settings that make the firmware listen *only* to the clock (for example a MIDI IN mode called CLOCK) as they are: they make it ignore incoming notes, so your MIDI keyboard would stop playing it.
- Some firmwares go back to their own tempo when the clock stops; they lock to it again as soon as it returns. FM-1 emulator keeps the clock running whenever the DAW has a tempo, so this does not happen while you work.
- If a firmware has no clock or sync setting, it cannot follow MIDI clock: set its tempo by hand to the DAW's tempo.
- Example, SLOOP: GLO, then SYSTEM, then SYNC = TRS.

8. MIDI Mapping the Knobs and Buttons

Every knob and button on the panel is also a host parameter, so you can control it from a MIDI controller or automate it: SELECT, PRESETS, ALGORITHM, KNOB 1-4 (group Knobs), and OCT-, OCT+, FX, SCL, ENV, LFO, EDIT, GLO, HOME, SAVE, ARP, SEQ, PLAY/STOP and REC (group Buttons). MASTER was already a parameter.

In Ableton Live:

- Open the device's parameter view: click the small triangle in the FM-1 emulator device's title bar to unfold its parameters (if a parameter is missing, click **Configure** and choose it).
- Press **Cmd+M** (Mac) or **Ctrl+M** (Windows) for MIDI Map Mode.
- Click the parameter, then move the knob or press the button on your controller.
- Press Cmd+M / Ctrl+M again to leave MIDI Map Mode.

How they behave:

- The FM-1's knobs are endless encoders. A mapped knob turns its encoder by the amount you move it: a full sweep of your controller knob is 128 clicks, one click for each of its 128 MIDI values, so settings with up to 128 steps (like a filter cutoff) cover their whole range. Settings with more steps than that cover part of their range from the controller: fine-tune the rest on the panel with the mouse.
- A mapped button presses the panel button once each time its value changes: each press of a pad or button on your controller is one press. Holding a button down is done on the panel with the mouse.
- Knobs turned from MIDI turn on the panel too, and turning a knob on the panel with the mouse moves its parameter, so Ableton's Configure mode picks it up and automation can record it.

9. Troubleshooting

- **No sound:** load a firmware first.
- **Panel keys play but the MIDI keyboard doesn't:** set MIDI channel to the firmware's channel (try 2, then 1). In Ableton, arm the track or set Monitor to In.
- **Not in time with the DAW:** check Sync to host is on and set the firmware's clock/sync setting to TRS (see chapter 7).
- **Crackles or "frames late":** raise Lead, lower CPU clock, or increase your DAW's buffer size.
- **Plugin not listed:** restart the DAW and rescan plugins. On Windows the VST3 needs FM-1.clap in Common Files\CLAP.
- **"The firmware stopped":** the firmware used an instruction the emulator doesn't know yet. Try another firmware.

10. Credits, Licence and Contact

FM-1 emulator by Transistor Labs. Emulation core: fm1-emu by Joakim Eriksson, MIT licence. Built with nih-plug (ISC), clap-wrapper and the CLAP SDK (MIT), Apple's AudioUnitSDK (Apache-2.0) and Steinberg's VST3 SDK (MIT). Firmwares belong to their authors. M-VAVE and FM-1 are names of their owner; this is an unofficial emulator.

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